

Shifter

Goal

Prepare IP for shifting with rotation (left or right) input data. This IP should receive N 2048 bytes, shift all received bits and send shifted packed.

Provided module

You can edit only the *task_1.sv* file. If you wish, you may add other files for supportive submodules.

The module has the following inputs and outputs:

Table 1 List of inputs and outputs of the task_1 module.

Signal name	Signal type	Bit length
i_clk	Input	1 bit
i_valid	Input	1 bit
i_rst	Input	1 bit
i_data	Input	8 bits
i_last	Input	1 bit
i_first	Input	1 bit
o_data	Output	8 bits
o_valid	Output	1 bit
o_last	Output	1 bit

The reset is synchronous and active high.

Input and output interfaces

In one packet task receives **from 2+1 to 2+2048 samples** of input data, 8 bits each. The last sample in a packet is indicated by signal *i_last*. The task should send 1 to 2048 of output samples, 8 bits each, and should indicate the last sample in the output packet by providing signal *o_last*.

Task receives data in the way shown in the figure below:

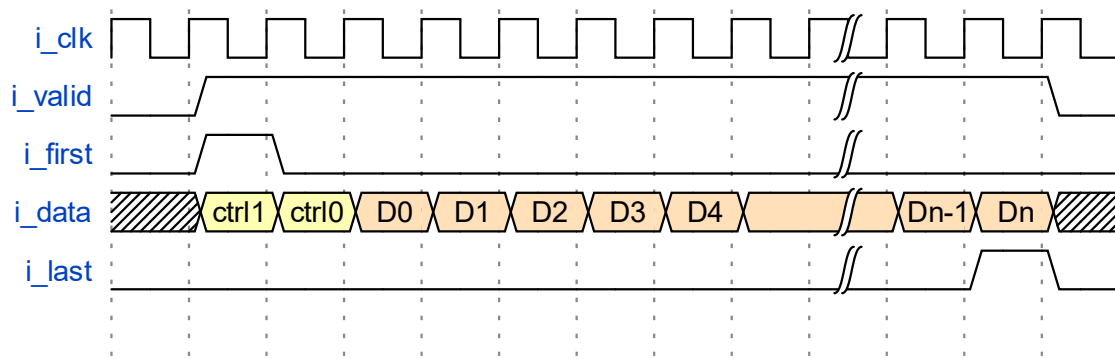


Figure 1 Input waveform.

To send data from the task in a proper way one needs to provide the *o_valid* signal that is synchronized with data on the *o_data* bus and *o_last* signal that is synchronized with the last sample in output packet.

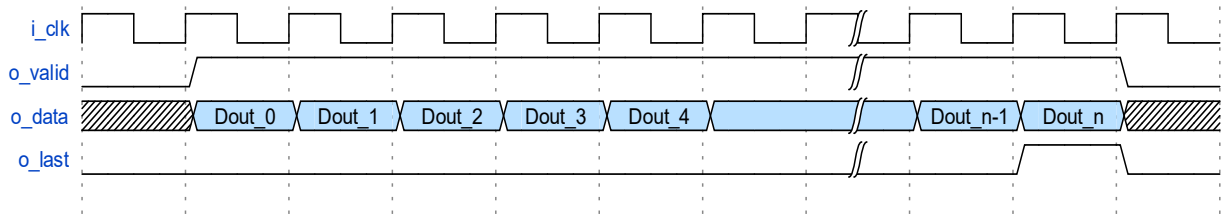


Figure 2 Output waveform

Output data can be sent as a continuous stream, or it can have regular or irregular gaps. The only requirement is providing *o_valid* and *o_last* signals.

Evaluating the task

IP should receive data packet, next shifted this packet left or right or rotated it also left or right. Next there should be send shifted (or rotated) data to the judge.

Two first received byte define behavior of IP:

ctrl_1 (7) – S/R

ctrl_1 (6) – L/R

ctrl_1 (5 : 0) – shift_value (13 : 8)

ctrl_0 (7 : 0) – shift_value (7 : 0)

S/R – 0 – shift packet, 1 – rotate packet

L/R – 0 – left, 1 – right

shift_value – define how many bits should be shifted

Example 1:

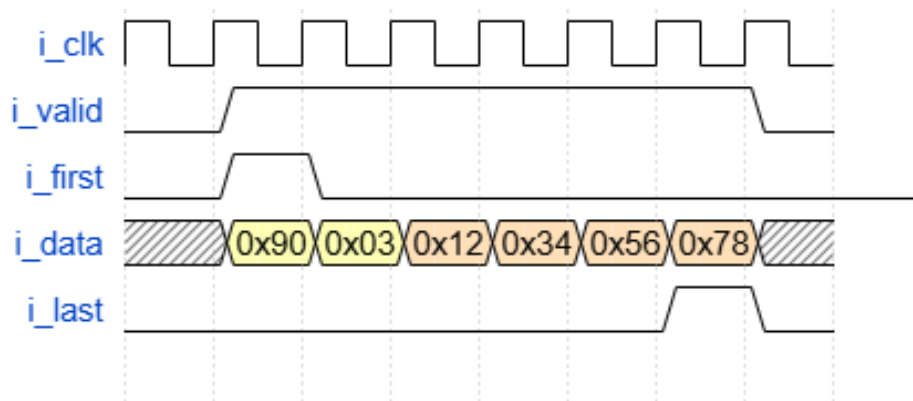


Figure 3 Input packet

Control - rotate left

Data packet - 0x12, 0x34, 0x56, 0x78

Output packet – 0x91, 0xA2, 0xB3, 0xC0

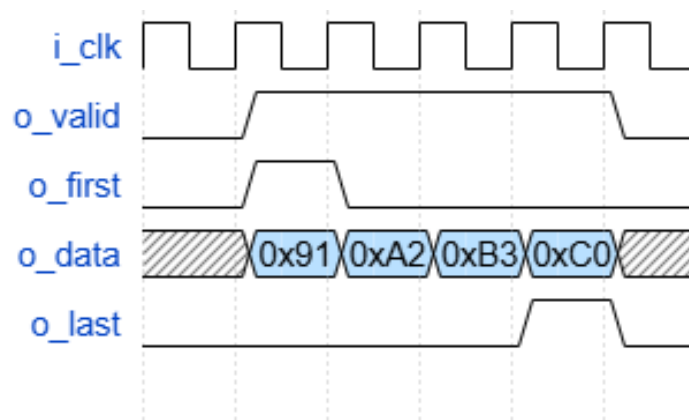


Figure 4 Output packet

About Development mode and Evaluation mode check chapter 2.3.1 **Testing modes** in the **FPGA_Hackathon_2025_Project_Guide** document.

Task will be tested with 4 gold vectors and pointed 2 points per vector.

Additional 5 points may be awarded for best **resource utilization** if the latency does not exceed 20 μ s. For more information, check chapter 2.3.2 **Bonus Points** in the **FPGA_Hackathon_2026_Project_Guide** document).